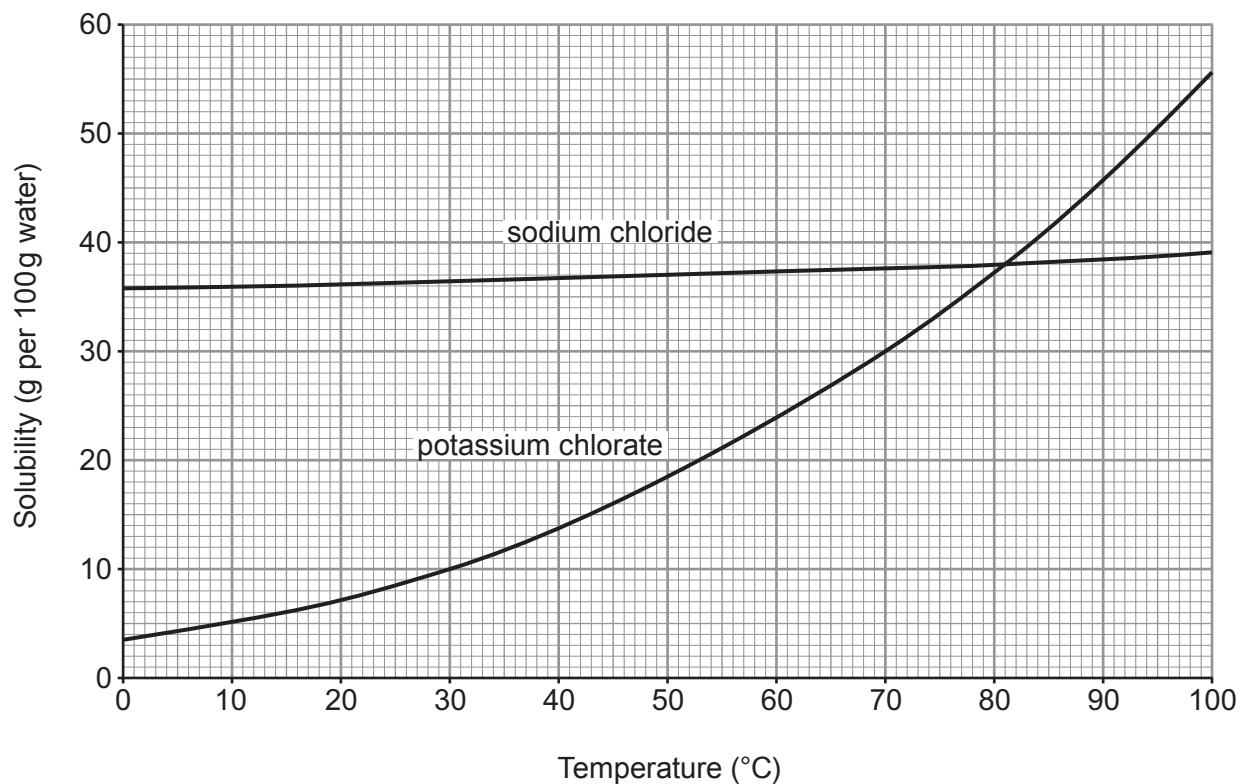


5. (a) The graphs below show the solubilities of sodium chloride and potassium chlorate in water at different temperatures.



- (i) State the temperature at which the two compounds have the same solubility. [1]

..... °C

- (ii) Calculate the mass of solid potassium chlorate that forms when a saturated solution in 100 g of water at 70 °C cools to 30 °C. [2]

mass = ..... g

- (b) (i) Calculate the relative formula mass ( $M_r$ ) of potassium chlorate,  $\text{KClO}_3$ .

[2]

relative formula mass = .....

- (ii) Calculate the percentage by mass of potassium in potassium chlorate.

[2]

percentage by mass = ..... %

- (c) Potassium chlorate forms potassium chloride and oxygen on heating.

Balance the symbol equation that represents the reaction taking place.

[1]

